

Integer Lunar Lander: Exact Landing Time

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Abstract

We prove the exact landing-time formula for the Integer Lunar Lander game. From state (i, j) , with velocity i and altitude j , the legal moves are

$$(i, j) \mapsto (i + 1, j + i + 1)$$

$$(i, j) \mapsto (i, j + i)$$

$$(i, j) \mapsto (i - 1, j + i - 1)$$

Let $T(i, j)$ be the minimum number of moves needed to reach $(0, 0)$ from $(i, j) \in \mathbb{Z}_{\geq 0}^2$. Define

$$C(q) = \begin{cases} 0 & q = 0 \\ \lceil 2\sqrt{q} \rceil & q > 0 \end{cases}$$

Then

$$T(i, j) = i + C\left(j + \binom{i}{2}\right)$$

Equivalently, with $\Delta_k = k(k + 1)/2$, for $i > 0$,

$$T(i, j) = i + C(j + \Delta_{i-1})$$

1 The $T(i, j)$ Function

This is a study into interesting properties and relationships generated by the minimum landing time in the Integer Lunar Lander game, which I call the $T(i, j)$ function.

Introduction

A position in the Integer Lunar Lander game is given by an ordered pair

$$(i, j)$$

where i is the velocity and j is the altitude. From the state (i, j) , the legal moves are

$$(i, j) \mapsto (i + 1, j + i + 1)$$

$$(i, j) \mapsto (i, j + i)$$

$$(i, j) \mapsto (i - 1, j + i - 1)$$

The aim is to reach

$$(0, 0)$$

in the smallest possible number of moves. We define this minimum number of moves as

$$T(i, j)$$

where $i, j \in \mathbb{Z}_{\geq 0}$.

The main formula is surprisingly simple. Define

$$C(q) = \begin{cases} 0 & q = 0 \\ q > 0 \end{cases} \quad \lceil 2\sqrt{q} \rceil$$

Then

$$T(i, j) = i + C\left(j + \binom{i}{2}\right)$$

Equivalently, using the triangular numbers

$$\Delta_k = \frac{k(k+1)}{2}$$

we have, for $i > 0$

$$T(i, j) = i + C(j + \Delta_{i-1})$$

This identity is interesting because it says that the i -th row of the game is just the zeroth row shifted by a triangular number and then increased by i .

Elementary Results Involving $T(i, j)$

This section contains a few elementary observations about the game. The key idea is that velocity can be removed first, and the remaining problem becomes an area problem.

Starting from velocity i , the quickest possible way to reach velocity 0 takes at least i moves

Proof. In one move, the velocity can change by at most 1. Therefore to get from velocity i to velocity 0, at least i moves are needed.

This lower bound is achieved by choosing acceleration -1 each time. The velocities after each move are

$$i-1, i-2, \dots, 1, 0$$

So the quickest possible way to remove velocity is just to decrease it by 1 at each step. $\square$

If the velocity is reduced from i to 0 as quickly as possible, then the altitude increases by

$$\binom{i}{2}$$

Proof. Choosing acceleration -1 at each step gives the velocities

$$i-1, i-2, \dots, 1, 0$$

The total altitude change during these i moves is therefore

$$(i-1) + (i-2) + \dots + 1 + 0$$

which simplifies to

$$\binom{i}{2}$$

So the state

$$(i, j)$$

naturally reduces to

$$\left(0, j + \binom{i}{2}\right)$$

□

This suggests that the real problem is the rest-start case. Namely, once the velocity is 0, how many moves are needed to land from altitude j ? This becomes an integer area problem.

Bounded-Step Areas

Consider integer paths

$$x_0, x_1, \dots, x_n$$

such that

$$x_0 = 0$$

$$x_n = 0$$

and

$$|x_t - x_{t-1}| \leq 1$$

These paths start at 0, end at 0, and change by at most one at every step. The largest nonnegative path of length n is the path:

$$0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow \left\lfloor \frac{n}{2} \right\rfloor \rightarrow \dots \rightarrow 2 \rightarrow 1 \rightarrow 0$$

The area under this path is

$$\left\lfloor \frac{n^2}{4} \right\rfloor$$

This turns out to be the exact largest possible area. Let

$$M_n = \left\lfloor \frac{n^2}{4} \right\rfloor$$

Any integer path satisfying

$$x_0 = 0$$

$$x_n = 0$$

and

$$|x_t - x_{t-1}| \leq 1$$

has

$$\sum_{t=1}^n x_t \leq M_n$$

Also, for every integer A satisfying

$$0 \leq A \leq M_n$$

there is a nonnegative integer path satisfying

$$x_0 = 0$$

$$x_n = 0$$

$$x_t \in \mathbb{Z}^+ \geq 0$$

$$|x_t - x_{t-1}| \leq 1$$

and

$$\sum_{t=1}^n x_t = A$$

Proof. For the upper bound, the path can rise by at most one from the left. So

$$x_t \leq t$$

It must also be able to return to zero from the right. So

$$x_t \leq n - t$$

Therefore

$$x_t \leq \min(t, n - t)$$

and hence

$$\sum_{t=1}^n x_t \leq \sum_{t=1}^n \min(t, n - t)$$

But the sum on the right is just the area of the tent path. Thus

$$\sum_{t=1}^n x_t \leq \left\lfloor \frac{n^2}{4} \right\rfloor$$

Now we show that every smaller nonnegative area also occurs. We use induction on n . The cases $n = 0$ and $n = 1$ are immediate. Suppose the result is true for length $n - 1$. Appending a final zero to any path of length $n - 1$ gives a path of length n with the same area, so every area up to

$$M_{n-1}$$

already occurs.

Only the new top areas remain. Let

$$p = \left\lfloor \frac{n}{2} \right\rfloor$$

Then

$$M_n - M_{n-1} = p$$

So the only areas left to construct are

$$M_n - s$$

where

$$0 \leq s \leq p - 1$$

Start with the tent path

$$c_t = \min(t, n - t)$$

which has area M_n . To get area $M_n - s$, shave s unit squares from the middle.

If $n = 2p$, lower by one the entries

$$t = p - s + 1 \text{ through } p$$

If $n = 2p + 1$, lower by one the entries

$$t = p - s + 2 \text{ through } p + 1$$

For $s = 0$, do nothing. The modified path is still nonnegative, still starts and ends at zero, and still has step size at most one. Its area is

$$M_n - s$$

Therefore every integer area from 0 to M_n occurs. □

The Rest-Start Case

Now consider the case where the lander starts with velocity 0. We want to compute

$$T(0, j)$$

For $j \in \mathbb{Z}^* \geq 0$,

$$T(0, j) = C(j)$$

Equivalently

$$T(0, 0) = 0$$

and for $j > 0$

$$T(0, j) = \left\lceil 2\sqrt{j} \right\rceil$$

Proof. Consider a landing trajectory of length n from

$$(0, j)$$

Let v_t be the velocity after t moves. Since we start and end at velocity zero,

$$v_0 = 0$$

$$v_n = 0$$

and since velocity changes by at most one per move,

$$|v_t - v_{t-1}| \leq 1$$

Landing means that the total altitude change is $-j$, so

$$j + \sum_{t=1}^n v_t = 0$$

Now set

$$x_t = -v_t$$

Then

$$x_0 = 0$$

$$x_n = 0$$

$$|x_t - x_{t-1}| \leq 1$$

and

$$\sum_{t=1}^n x_t = j$$

So landing from rest at altitude j is exactly the same as finding a bounded-step path with area j .

By the previous proposition, a length n path can realize area j if and only if

$$\left\lfloor \frac{n^2}{4} \right\rfloor \geq j$$

The least such n is 0 when $j = 0$, and for $j > 0$ it is

$$\left\lceil 2\sqrt{j} \right\rceil$$

Therefore

$$T(0, j) = C(j)$$

□

The Main Formula

We now prove the exact landing-time formula. The important quantity is not just j , but rather

$$j + \binom{i}{2}$$

This is the altitude after the fastest possible velocity-removal phase. For all $i, j \in \mathbb{Z}^* \geq 0$,

$$T(i, j) = i + C\left(j + \binom{i}{2}\right)$$

Proof. We first prove the lower bound. Suppose there is a landing trajectory from

$$(i, j)$$

to

$$(0, 0)$$

of length n . Since velocity can change by at most one per move, we must have

$$n \geq i$$

Let

$$m = n - i$$

Let v_t be the velocity after t moves, and define

$$x_t = -v_t$$

Then

$$x_0 = -i$$

$$x_n = 0$$

$$|x_t - x * t - 1| \leq 1$$

and the landing condition becomes

$$\sum_{t=1}^n x_t = j$$

From the left, the path cannot rise faster than one per step, so

$$x_t \leq t - i$$

From the right, it must be able to return to zero by time n , so

$$x_t \leq n - t$$

Therefore

$$x_t \leq \min(t - i, n - t)$$

and so

$$j \leq \sum_{t=1}^n \min(t - i, n - t)$$

Now let

$$r = t - i$$

Then r runs from $1 - i$ to m , and the sum becomes

$$\sum_{r=1-i}^m \min(r, m - r)$$

The negative part contributes

$$\sum_{r=1-i}^{-1} r = -\binom{i}{2}$$

and the nonnegative part is just the tent area

$$\sum_{r=0}^m \min(r, m - r) = \left\lfloor \frac{m^2}{4} \right\rfloor$$

Therefore

$$j \leq \left\lfloor \frac{m^2}{4} \right\rfloor - \binom{i}{2}$$

or equivalently

$$j + \binom{i}{2} \leq \left\lfloor \frac{m^2}{4} \right\rfloor$$

By the rest-start formula, the least such m is

$$C\left(j + \binom{i}{2}\right)$$

Therefore

$$n \geq i + C\left(j + \binom{i}{2}\right)$$

Now we prove the upper bound. Reduce the velocity from i to 0 as quickly as possible. This takes i moves and gives the velocity sequence

$$i-1, i-2, \dots, 1, 0$$

During these moves, the altitude increases by

$$\binom{i}{2}$$

So the state becomes

$$\left(0, j + \binom{i}{2}\right)$$

By the rest-start formula, this state can be landed in

$$C\left(j + \binom{i}{2}\right)$$

more moves. Hence

$$T(i, j) \leq i + C\left(j + \binom{i}{2}\right)$$

Together with the lower bound, we get

$$T(i, j) = i + C\left(j + \binom{i}{2}\right)$$

□

Relation To The OEIS Conjecture

OEIS A360923 records this Integer Lunar Lander game and gives a conjectured row-shift formula. Let

$$\Delta_k = \frac{k(k+1)}{2}$$

The conjectured relation is

$$T(i, j) = T(0, j + \Delta_{i-1}) + i$$

for $i > 0$. Since

$$\Delta_{i-1} = \frac{i(i-1)}{2} = \binom{i}{2}$$

the main formula gives

$$T(i, j) = i + C(j + \Delta_{i-1})$$

But the rest-start formula gives

$$T(0, j + \Delta_{i-1}) = C(j + \Delta_{i-1})$$

Therefore

$$T(i, j) = T(0, j + \Delta_{i-1}) + i$$

which proves the row-shift formula.

The zeroth row is also simplified. For $j > 0$, OEIS writes the zeroth-row formula as

$$T(0, j) = 1 + \left\lfloor \sqrt{4j - 3} \right\rfloor$$

But the rest-start result gives

$$T(0, j) = \left\lceil 2\sqrt{j} \right\rceil$$

These two expressions are equal for every $j > 0$. The case $j = 0$ is separate, since

$$T(0, 0) = 0$$

So the full formula can be written as

$$T(i, j) = i + \begin{cases} 0 & j + \binom{i}{2} = 0 \\ j + \binom{i}{2} > 0 \end{cases} \left\lceil 2\sqrt{j + \binom{i}{2}} \right\rceil$$

No-Crash Optimal

The proof above also suggests a stronger result. Although the original game allows negative altitudes, we can still choose an optimal trajectory that never goes below altitude zero. For every $i, j \in \mathbb{Z}_{\geq 0}$, there is an optimal trajectory from

$$(i, j)$$

to

$$(0, 0)$$

whose altitude is never negative

Proof. Use the optimal strategy from the main formula. First reduce velocity from i to 0 as quickly as possible. During this phase the velocities are

$$i - 1, i - 2, \dots, 1, 0$$

These are all nonnegative, so the altitude never decreases during this phase.

After this, the state is

$$\left(0, j + \binom{i}{2}\right)$$

Let

$$q = j + \binom{i}{2}$$

If $q = 0$, we are done. If $q > 0$, the bounded-step area result gives a nonnegative path of length

$$C(q)$$

such that

$$x_0 = 0$$

$$x_{C(q)} = 0$$

$$x_t \in \mathbb{Z}^* \geq 0$$

$$|x_t - x * t - 1| \leq 1$$

and

$$\sum x_t = q$$

Now use velocities

$$v_t = -x_t$$

during the second phase. After t moves in this phase, the altitude is

$$q - \sum_{s=1}^t x_s$$

Since the partial sums never exceed the total area q , the altitude never becomes negative. The total number of moves is

$$i + C\left(j + \binom{i}{2}\right)$$

which is optimal. □

Conjecture

The formula

$$T(i, j) = i + C\left(j + \binom{i}{2}\right)$$

completely solves the two-dimensional nonnegative starting case. However, the structure of the table still seems to contain further patterns.

For example, OEIS A360923 also suggests that nearby entries tend to differ by only a small amount. Since

$$C(q) = \lceil 2\sqrt{q} \rceil$$

for $q > 0$, the growth is controlled by square-root jumps. This makes it natural to study the difference arrays

$$T(i+1, j) - T(i, j)$$

and

$$T(i, j+1) - T(i, j)$$

as well